Cross-Modal Consistency Is Not a Cross-Generator Invariant: An Attributed Negative-Result Study of Lightweight Frozen-Backbone Deepfake Detection

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***Abstract*—Native joint audio-video generators (Seedance 2.0, SkyReels-V4, Sora 2) now synthesise tightly synchronised speech and imagery, erasing the pixel-level traces and gross inter-modal lag that earlier deepfake detectors exploit. A widely shared hypothesis holds that *cross-modal consistency* is a generator-invariant cue: real speech and lip motion are coupled by physiology, so any forgery must partially violate that coupling. We falsify this hypothesis in a precisely scoped, resource-constrained setting. We build Sync-X, a lightweight single-GPU detector that couples two *frozen* self-supervised backbones (CLIP ViT-B/16 for vision, wav2vec2-base-960h for audio) through cross-attention, articulatory-probe, and dynamic-fusion adapters. Sync-X reaches an in-domain validation AUC of 0.9758 on DeepSpeak v1.1 yet collapses on held-out cross-domain benchmarks (FakeAVCeleb 0.6187, Deepfake-Eval-2024 0.5350, HiFi-AVDF 0.4735). A single-variable diagnostic programme yields five independent lines of evidence: under offline-frozen, single-source supervision, audio-visual consistency never becomes a cross-generator invariant. Each convenient signature is a training-domain fingerprint that reverses or vanishes across generators, and a linear centred-kernel-alignment (CKA) carrier probe returns *NOT_SEPARABLE*. One bounded positive result survives: a long-term temporal consistency adapter (LTCA) is the only recipe to push Deepfake-Eval-2024 above 0.55 (AUC 0.5796). We contribute an attributed negative result with its correctly bounded positive, plus a reusable single-variable attribution methodology.**

***Index Terms*—deepfake detection, audio-visual consistency, cross-modal learning, domain generalisation, negative result, frozen backbone, self-supervised learning**

I. Introduction

Modern generative models can synthesise an entire talking-head scene, audio included, from a single text prompt. Detectors built for the 2017–2023 GAN/autoencoder era (MesoNet [1], LipForensics [2]) exploited the stitched artefact trail of those methods (irregular blinking, boundary ghosting, gross lip–audio lag). Modern joint-generation systems such as Seedance (1.5 Pro, 2.0) [3], [4], SkyReels-V4 [5], Sora 2, Veo 3, Kling, WAN generate audio and video in one unified architecture with near-perfect synchronisation, so those artefact-centric detectors degrade sharply. The collapse is quantitative: on Deepfake-Eval-2024, an in-the-wild benchmark, the best open detector reaches only AUC 0.58 [6].

The open question is which signal survives once spatial artefacts and gross desynchronisation are eliminated. The *consistency-first* hypothesis argues that cross-modal consistency is the answer: because articulation physically constrains the acoustic signal and vice versa [7], any forgery must partially violate that coupling, so consistency could be a generator-invariant cue. Detectors built on this idea (MDS [8], AVFF [9], HAVIC [10], X-AVDT [11]) outperform artefact-centric baselines, and HAVIC reports cross-dataset gains.

We test it under a constrained single-GPU paradigm: two *frozen, separately pre-trained* self-supervised backbones (CLIP ViT-B/16, wav2vec2-base-960h) with only light adapters trained. We do *not* co-train the heavy backbones, and we do *not* access generator internals: both are out of reach of this setting and both are required by the strongest methods (HAVIC, X-AVDT). The resulting failure mode is one the success-framed literature rarely reports.

a) Research questions.: We frame the study as falsification questions:

- **RQ1:** Does a lightweight, frozen-backbone multimodal framework achieve *cross-domain* robustness rather than in-domain accuracy alone?
- **RQ2:** Where does the cross-domain bottleneck lie: in supervision, modality budget, or the frozen representation itself?
- **RQ3:** Is audio-visual consistency, as captured here, a stable cross-generator invariant, or a generator-specific fingerprint that reverses or vanishes across generators?

b) Contributions.: (C1) A negative result with a precise mechanism: under the offline-frozen, single-source framework, consistency does not become a cross-generator invariant; five independent evidence lines converge on the root cause, showing that what is falsified is the *feature framework*, not the consistency direction. (C2) A bounded, correctly attributed positive: the long-term temporal consistency adapter (LTCA) is the only recipe to push Deepfake-Eval-2024 above 0.55 (0.5796). (C3) A reusable single-variable attribution methodology (shortcut scan → modality budget → carrier probe).

c) Terminology.: Consistency denotes cross-modal articulatory (lip–audio) correspondence and its temporal regularity, not naive temporal lag. A *cross-generator invariant* is a signal whose real-vs-fake direction and strength are stable across generators; no candidate examined here qualifies.

d) Why report a negative result.: A null result is a contribution when it rules out a hypothesis the field expects to hold, and does so with a mechanism. The consistency-first hypothesis is such a hypothesis: it underlies several recent systems and is the implicit consensus of the multimodal-deepfake literature, yet it fails here in the regime of frozen separate backbones and a single training source. What we contribute is the localisation of that failure, together with one bounded success (LTCA).

II. Related Work

We summarise only the strands that bound our expectations; the complete literature review appears in Part 1 of this thesis and is not repeated here.

a) Artefact-centric and lip-sync detectors.: The first generation detected spatial, temporal or stitching artefacts in single frames. FaceForensics++ [12] and MesoNet [1] set the closed-benchmark baselines, and lip-aware detectors (LipForensics [2], landmark-constancy attributes [13], and the Wav2Lip synthesis family [14] used as a stress test) reach high accuracy in-domain by exploiting exactly the micro-mismatch that native joint generators now erase. They remain strong on their training generators but rely on generator-specific traces.

b) Cross-modal consistency detectors and joint backbones.: Consistency-based detection scores audio–visual correspondence directly: MDS [8] measures synchronisation dissonance, AVFF [9] learns correspondence through complementary masking, and audio-visual contrastive pretraining [15], [16] injects correspondence into the backbone. The most transferable systems co-train a *joint* representation with an alignment objective: AV-Lip-Sync+ [17] couples AV-HuBERT with a multi-scale temporal convolutional network, HAVIC [10] pre-trains audio-visual coherence priors on authentic videos, and X-AVDT [11] reads generator-internal cross-attention via DDIM inversion. All three require heavy co-training and/or white-box access; our study deliberately freezes both backbones to test whether the consistency cue survives the *offline, single-source* setting a lightweight detector can actually afford.

c) Generalisation and shortcut failure modes.: Detectors trained on known generators learn generator-specific fingerprints and collapse on unseen generators [6], [18]. Post-hoc analyses refine this: existing forgeries share a small family of universal artefacts [19]; shortcut cues such as leading silence inflate early audio-visual benchmarks, notably FakeAVCeleb and AV-Deepfake1M [20], [21]; and frequency-domain consistency supplies a complementary axis that standard AV features miss [22]. Our study occupies the intersection: does the nominally physical, domain-agnostic consistency cue transfer across generators under a frozen, single-source design, or does it degrade into a training-bound fingerprint?

d) Evaluation benchmarks.: Cross-domain evidence is drawn from three benchmarks that never see training (Deepfake-Eval-2024 [6], FakeAVCeleb [23], and HiFi-AVDF [10]) with multimodal-benchmark analyses [24] and the excluded TalkingHeadBench [25] and MVAD [26] as context; scope selection is justified in Section IV.

III. Method: The Sync-X Framework

Sync-X is the framework under test: its design encodes the hypothesis that frozen separate backbones suffice and that consistency and lip biomechanics are discriminative. Section V tests that hypothesis.

A. Offline Feature Pre-extraction

To decouple expensive encoding from training, each clip is processed once: MediaPipe FaceMesh [27] yields 64 perioral landmarks per frame ($v_{lip} \in \mathbb{R}^{T_v \times 192}$) and a 224×224 face crop encoded by the frozen CLIP ViT-B/16 [28], [29] into 512-D per-frame embeddings; audio at 16 kHz is encoded by the frozen wav2vec2-base-960h [30] into $A \in \mathbb{R}^{T_a \times 768}$; a 112×112 grayscale lip ROI supplies a visual prior. Clips are resampled to 100 visual / 200 audio frames (≈4 s), then to the model length $T = 50$, and all tensors are cached in half precision. Caching once reduces per-epoch time from hours to minutes.

B. Architecture

Let $A = E_a(x_a) \in \mathbb{R}^{T_a \times 768}$, $V = E_v(x_v) \in \mathbb{R}^{T_v \times 512}$, with lip crops C and landmarks P.

a) LCMA (bidirectional cross-modal attention).: Both streams are projected to a shared width $d = 256$ and resampled to $T = 50$: $\tilde{A} = R_T(AW_a)$, $\tilde{V} = R_T(VW_v)$. Two bidirectional cross-attention blocks align and fuse them, capturing intrinsic audio-visual correspondence rather than uni-modal artefacts. The pooled output is $h_{lcma} \in \mathbb{R}^{512}$.

b) LACA (dual-signal lip probe).: The perioral landmark sequence is turned by a Savitzky–Golay frontend and a compact MLP + bidirectional GRU into a per-frame representation K. Two orthogonal sub-branches consume K: (i) a biomechanical probe mean-pools K and projects to 512-D to expose non-physiological jitter; (ii) a lip–audio correspondence probe compares K against the projected wav2vec representation via $|K-a|$, feeding an MLP to flag articulatory–acoustic mismatch. The two 512-D vectors are fused into h_{laca}.

c) Dynamic Weight Fusion (DWF) and head.: DWF learns sample-specific branch weights $h = \alpha_\ell h_{lcma} + \alpha_p h_{laca}$. The classifier is an Additive Angular Margin (AAM) softmax head in cosine form (CosFace-style [31], [32]) with margin $m = 0.30$, scale $s = 30$ (fp32), chosen for the binary task and the frozen feature distribution.

d) Fusion and pooling in detail.: The two branch vectors enter the Dynamic Weight Fusion after independent normalisation, and a small two-layer gate produces the per-sample weights α_ℓ, α_p through a softmax, so the head can lean on whichever stream a given clip demands. The pooled representation h is rescaled to unit norm before the AAM head, matching the cosine-form classifier. All trainable parameters live in the projections, the gate, and the head; the encoders contribute no gradients, which is what makes the single-variable discipline precise and keeps memory flat as benchmarks grow. Weighting rather than concatenation is a design choice: it

lets the gate respond to each clip separately, and it exposes the two weights for inspection. We do not ablate it against concatenation, so we treat it as a design decision rather than an established advantage.

C. Training Objectives

Stage 1 (self-supervised, authentic VoxCeleb2 clips only, 60 epochs) optimises InfoNCE on LCMA projections, a kinematic pseudo-jitter loss on the LACA biomechanical branch, and a shared-space lip–audio alignment term. Stage 2 (supervised real/fake, 80 epochs) uses

$$\mathcal{L}_{\text{stage2}} = \mathcal{L}_{\text{AAM}} + \lambda_c \mathcal{L}_{\text{corr_lcma}} + \lambda_s \mathcal{L}_{\text{shared}} + \lambda_{\text{al}} \mathcal{L}_{\text{align}} + \lambda_u \mathcal{L}_{\text{uniform}}$$

with the principal weights $\lambda_c = 0.30$, $\lambda_s = 0.10$, $\lambda_{\text{al}} = 0.10$, $\lambda_u = 0.50$. $\mathcal{L}_{\text{align}}$ and $\mathcal{L}_{\text{uniform}}$ are the hyperspherical alignment/uniformity regularisers of GenD [33]; same-source real/fake pairing is exploited through batch sampling. All heavy backbones remain frozen in both stages. Table I lists the configuration.

a) Choice of supervision and margin.: The Stage-2 head is a CosFace-style Additive Angular Margin softmax rather than plain binary logistic, with margin $m = 0.30$ and scale $s = 30$ held in fp32. Over a frozen feature distribution the margin sharpens the angular decision boundary between real and fake, while a fixed scale keeps the logit range commensurate with the geometry of that space. We fix m and s as constants and never re-tune them per benchmark, so that cross-domain numbers reflect the method rather than per-domain tuning; tuning on a held-out benchmark would leak signal across the domain boundary being measured. The single-variable protocol never swaps a hyperparameter between recipes for the same reason.

The design deliberately keeps the backbones frozen and both stages shallow. Freezing makes every experiment cheap (a single cloud GPU), makes the ablations single-variable in a precise sense (only the light trainable head and objective change, never the encoders) and forces any cross-domain signal to be carried by the frozen representations themselves rather than by capacity that co-adapts to an in-domain bias. Stage 1 is included so that the audio–visual stream learns an intrinsic-coupled prior on authentic data before any real/fake supervision is introduced; this mirrors the consistency-first hypothesis at the representation level. The choices are minimal so that success or failure can be attributed to a specific component rather than to in-domain accuracy.

b) Why CLIP and wav2vec2.: The two encoders are chosen to represent the mainstream of frozen, black-box feature extraction rather than for any special property. CLIP ViT-B/16 is a public vision-language backbone whose face-tuned features already give a strong uni-modal baseline; wav2vec2-base-960h is the standard frozen speech encoder with a CTC-agnostic 768-D frame embedding. Both load as drop-in PyTorch modules, need no task-specific fine-tuning to run, and can be cached in half precision for a corpus of this size on one 24 GB GPU. For attribution, neither was trained with any audio–visual alignment objective: wav2vec2 learns single-modality speech self-supervision, and CLIP learns image–text contrast, not speech–lip correspondence. They are therefore a clean test bed for the offline hypothesis: if two backbones with effectively *no* built-in audio–visual coupling expose no transferable consistency signal, the shortfall attaches to the feature framework rather than to one encoder.

IV. Experimental Setup

a) Datasets.: Sync-X is trained on **DeepSpeak v1.1** [34] (face-swap, lip-sync, voice-cloned audio; identity-disjoint split; real and fake clips from the same source recordings, enabling GenD's same-source pairing). VoxCeleb2 [35] supplies authentic clips for Stage-1 SSL only. Generalisation is evaluated *zero-shot* on three benchmarks never seen during training: **Deepfake-Eval-2024** [6] (the primary in-the-wild test), **FakeAVCeleb** [23] (including a documented leading-silence shortcut cue [21]), and **HiFi-AVDF** [10] (forgeries by six *commercial second-generation* generators: Veo 3.1, Kling 2.5, Seedance 1.0, PixVerse V5, WAN 2.5, Sora 2, entirely absent from training). TalkingHeadBench and MVAD are excluded as redundant with HiFi-AVDF. No benchmark contributes to both training and generalisation evidence; no generator appears in both training and cross-domain sets.

b) Benchmark composition.: Deepfake-Eval-2024 is a large in-the-wild multimodal set whose fake clips come from real-world forgeries rather than one laboratory engine, making it the harshest generalisation test and the primary number we report. FakeAVCeleb organises real and fake clips along two orthogonal axes (real/fake audio × real/fake video), giving four fine-grained classes (Real–Real, Fake Audio–Real Video, Real Audio–Fake Video, and Fake–Fake) and is so class-imbalanced (178 real vs. 19,547 fake) that its AUC is reported only for reference. HiFi-AVDF supplies forgeries by six *second-generation native-joint* commercial engines (Veo 3.1, Kling 2.5, Seedance 1.0, PixVerse V5, WAN 2.5, Sora 2) that synchronise speech to generated mouths in a single generation pass; because none of these appears in training, HiFi is the decisive set for whether consistency is generator-invariant, and its per-generator rows feed the carrier probe. VoxCeleb2 contributes only authentic audio–video pairs to Stage 1 self-supervision and is never used as a detection set.

c) Temporal alignment and evaluation hygiene.: Every benchmark is preprocessed identically so the frozen encoders receive the same 4-second centre window, cropped and resampled to the model length $T = 50$ (100 visual / 200 audio frames before temporal pooling): 224×224 face crops for CLIP, 112×112 grayscale lip regions, 64 perioral landmarks per frame from MediaPipe FaceMesh, and 16 kHz mono audio through a log-mel front-end. Frame and landmark statistics are consistent across DeepSpeak, FakeAVCeleb, DFE-2024 and HiFi-AVDF, so a cross-domain drop cannot be blamed on mismatched preprocessing. Each clip is encoded exactly once and the cached tensors are reused by every recipe and ablation, which is what makes a many-configuration comparison feasible on one GPU and keeps the deltas single-variable.

TABLE I
PRINCIPAL TRAINING CONFIGURATION.

Parameter	Stage 1	Stage 2
Optimiser	AdamW, LR 1×10^{-4}, wd 0.01, warmup 500	
Batch size	1024	512
Backbones	CLIP ViT-B/16 (512-D), wav2vec2-base-960h (768-D), frozen	
AAM	cosine form, $m = 0.30$, $s = 30$, fp32	
Temporal length T	50	
Loss weights	$\lambda_c = 0.30$, $\lambda_s = 0.10$, $\lambda_{\text{al}} = 0.10$, $\lambda_u = 0.50$	
Precision	AMP fp16 (AAM fp32), grad clip 1.0	

d) Metrics and gate.: The primary metric is video-clip-level AUC (imbalance-invariant). All reported primary results are single-seed (seed 0) measurements; standard deviations are not fabricated or extrapolated: they are omitted by design because the claims are qualitative/mechanism-level, and multi-seed variance is reported only where a numerical delta is at stake (the GenD ablation, which uses 3 seeds). We adopt a strict cross-domain gate: a cue qualifies as transferable only if its balanced AUC is stable in direction and ≥ 0.65; offline probes use a strength gate $\max(\text{auc}, 1 - \text{auc}) \geq 0.6$. FakeAVCeleb (severely imbalanced: 178 real vs. 19,547 fake) is reported as a reference, not as the primary basis for judgement.

e) Reproducibility, ethics and data privacy.: Full configuration, preprocessing code, analysis scripts and a reproducibility guide are provided in Part 3. All runs use a fixed software environment (PyTorch 2.8, CUDA 12.8, torchvision 0.23) on a single NVIDIA RTX 4090D (24 GB); preprocessing runs on CPU (MediaPipe XNNPACK) with memory limits, and training is a single-GPU job, making every number reproducible on one rented instance. This study uses only public, synthetic or licensed deepfake and speech corpora (DeepSpeak v1.1, VoxCeleb2, FakeAVCeleb, Deepfake-Eval-2024, HiFi-AVDF, FaceForensics++); no new data involving human participants was collected, no individuals are re-identified, and feature encodings are stored as numerical tensors without redistributing raw recordings. The work develops defensive detection technology from secondary analysis of publicly available data, and is accordingly a negligible-risk use of secondary public datasets in line with Monash University's low-risk ethics review process; no Monash ethics submission is required for this scope.

f) Single-variable protocol.: Every comparison changes one factor at a time on top of the principal configuration (Section V, GenD ablation).

g) Evaluation protocol and probe construction.: All numbers are produced by the same downstream evaluation harness so that deltas are comparable across recipes. In-domain validation AUC is measured on the identity-disjoint held-out partition of DeepSpeak v1.1 that shipped with the corpus; cross-domain AUC is measured zero-shot (no cross-domain clip is ever seen during training or validation). Checkpoints are selected by in-domain validation AUC with early stopping capped at 80 epochs, so a checkpoint cannot overfit to a held-out benchmark it never saw. Offline probes (the audio linear probe reporting 0.827, the physical/frequency proxies, and the carrier probe) are trained on cached, pre-extracted features with a held-out split internal to each probe, using the same strength gate $\max(\text{auc}, 1 - \text{auc}) \geq 0.6$ so that a probe must beat the negation of its own direction to count as carrying signal. The classifier and probe heads are deliberately tiny (a linear layer or a single hidden layer), so probe results reflect the information content of the frozen features rather than the capacity of a trained network.

h) Baselines.: HAVIC and X-AVDT are not run as baselines: both require co-trained backbones or white-box access to generator internals, which lie outside the frozen, black-box regime this study evaluates. We therefore report no head-to-head comparison, and our claims concern attribution rather than ranking.

V. Results and Discussion

We first fix how the numbers should be read, so that the negative result is not over-interpreted. AUC is used because it is imbalance-invariant for a binary real/fake task at clip level. A value near 0.5 is random; we treat ≥ 0.65 as the transfer-quality gate for a *stable-direction* cue, because below that a small change in test composition or seed flips the direction, and a direction that can flip is not a cross-generator invariant by definition. Magnitude matters for in-domain strength (0.9873 is unambiguous), but for cross-domain evidence the discriminator is the *sign* and its stability across generators: this is why the carrier probe reports strength $\max(\text{auc}, 1 - \text{auc})$ and why a single sign-reversed generator (veo3.1) already undermines the invariant claim. Throughout, *transfer* means a cue whose real-vs-fake direction and magnitude hold under the train/test shift, not merely a non-random AUC.

A. The In-Domain–Cross-Domain Collapse

Table II reports in-domain validation AUC and zero-shot AUC for the principal recipes.

Every recipe that carries an in-domain measurement saturates at ≥ 0.87 yet lands at or near random (0.44–0.58) on the held-out sets; HiFi-AVDF is consistently ≤ 0.51. This is *domain overfitting / shortcut learning*, not classical overfitting: in-domain validation stays high on the held-out identity-disjoint partition, so the pathology is specifically the non-transfer of generator-specific discriminative features. Figure 1

TABLE II

IN-DOMAIN (DEEPSPEAK V1.1) AND ZERO-SHOT CROSS-DOMAIN AUC OF THE PRINCIPAL RECIPES. `V2` = FULL FINE-TUNE; `FREEZE` = FREEZE ALL ADAPTERS EXCEPT HEAD (THIS RUN ALSO ENABLED MODALITY DROPOUT 0.3); `VIS-ONLY` = FREEZE THE VISUAL BRANCH; **LTCA** = + LONG-TERM TEMPORAL CONSISTENCY ADAPTER; **ALIGN** = LTCA + EXPLICIT END-TO-END ALIGNMENT; **HAVIC-RECON** = + RECONSTRUCTION CONSISTENCY PRIOR AT STAGE-1.

Recipe	In-dom. val	FakeAV	DFE-2024	HiFi-AVDF
v2 (full fine-tune)	0.9758	0.6187	0.5350	0.4735
freeze	0.8740	0.4373	0.4410	0.4592
vis-only	—	0.5189	0.5558	0.4975
LTCA	0.9462	0.6009	**0.5796**	0.4834
align (λ=0.1)	0.9565	0.5999	0.5351	0.5097
HAVIC-recon	—	0.5643	0.5081	0.4933

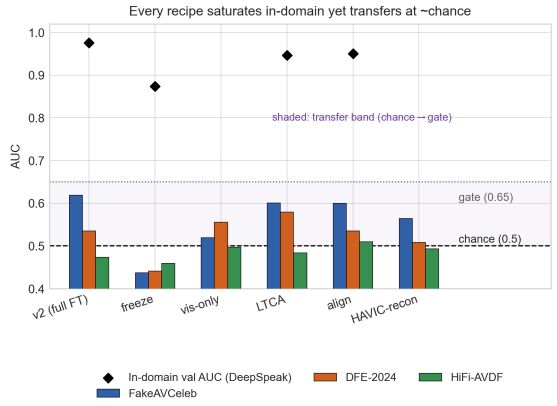


Fig. 1. In-domain validation AUC (black diamonds) versus zero-shot cross-domain AUC for all principal recipes. Cross-domain points stay at or near chance; no recipe exits the shaded transfer band.

makes the gap visual: in-domain points (diamonds) sit at 0.87–0.98 while every cross-domain bar stays inside the shaded transfer band (chance-to-gate).

B. Modality Budget: A Broken Audio-Supervision Wire

Zero-masking the v2 checkpoint on the in-domain validation set (video-only by zeroing audio, audio-only by zeroing visual) isolates the modal contribution: full 0.9758, video-only 0.9873, audio-only 0.5197 (random). The in-domain strength is carried almost entirely by a *visual, training-domain-bound* representation; the audio branch receives almost no direct real-vs-fake gradient. A linear probe on the *audio embeddings* reaches AUC 0.827: the audio *features contain* separable signal that the supervised adapter fails to exploit, so the bottleneck lies downstream of the frozen encoders, in the supervision/head, not the features.

C. Training-Recipe Axis: Three Consecutive Negatives

Freezing `lcma.*/laca.*` and training only the DWF/audio heads (3,586 trainable parameters) yields *dropped* cross-domain numbers (FakeAV 0.4373 below random, DFE-2024 0.441, HiFi 0.4592); in-domain plateaus at 0.874. This run also enabled modality dropout (0.3), so its numbers reflect the combined setting rather than freezing alone; the in-domain plateau at 0.874 nonetheless isolates the dominant factor. Freezing only the visual branch to force audio use gives near-random results (FakeAV 0.5189, DFE-2024 0.5558, HiFi 0.4975). Because all three recipes operate on the *same* offline frozen features and all fail, the bottleneck is the *frozen representation itself*, not the supervision/freeze strategy; this shifts attribution from the training axis to the feature axis.

D. Directional Probes: Towards a Partial Positive

a) Physical proxies fail to transfer.: Upgrading the LACA kinematics to explicit physical proxies (Savitzky–Golay jerk with moments) and measuring real-vs-fake strength shows they transfer nowhere: DFE-2024 and HiFi sit at ~0.5. The decisive evidence is *sign instability*: on the training domain the jerk Cohen's d is negative, but on FakeAVCeleb the sign flips positive, so the same proxy is positively discriminative on one generator family and negatively on another; a physical invariant cannot reverse direction across generators.

b) Frequency and coherence proxies fail the gate.: The core coherence signal (cross-spectral $\texttt{sync_corr}_{1\text{–}8\text{Hz}}$) is ~0.5 on *all* sets, including training: at this proxy granularity audio-visual coherence is not captured by the features. Jaw autocorrelation is strong in-domain (0.740) but reverses sign across domains (d = +0.98 train vs. −0.53 FakeAV); the audio-envelope autocorrelation is the only proxy above 0.6 cross-domain (0.649 on DFE-2024) but is single-modality smoothness, not cross-modal coherence. None meets the gate with stable direction. Figure 2 summarises the strength and sign of all physical and structural proxies across domains.

c) LTCA: the one partial positive, correctly bounded.: We implemented direction B as a third branch: a *long-term temporal consistency adapter* that consumes LCMA post-attention content embeddings through dilated temporal convolution ($\times 1, 2, 4$ + pooled MLP) and a cross-attention subsystem with an explicit temporal coherence scalar; the DWF gate is extended to three-way. LTCA pushes Deepfake-Eval-2024 to **0.5796** (the only recipe to break the 0.55 barrier) while FakeAV holds at 0.6009 and does not collapse. This is the strongest positive outcome: an attributable cross-domain gain on a large in-the-wild benchmark, produced by a lightweight adapter on frozen features.

The partial failure is attributed in full. HiFi-AVDF *is flat* under LTCA (0.4834 vs. 0.4735 v2). A sync-scalar probe and a `-fta_no_sync` ablation isolate the cause. First, the frame-wise sync scalar is near-random on DFE-2024 (0.508) yet

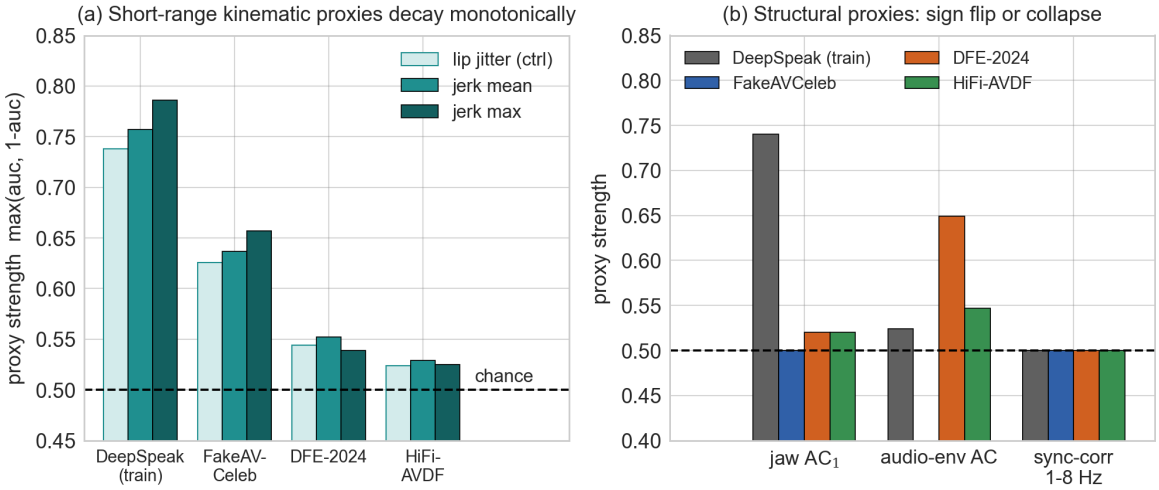


Fig. 2. Physical and structural proxies: strength $\max(\text{auc}, 1-\text{auc})$. (a) Short-range kinematic proxies decay from the training domain to held-out sets. (b) Structural proxies either flip sign across domains or stay at chance.

DFE-2024 gains the most, so DFE-2024's gain comes from the *long-horizon content backbone*, not the explicit sync signal. Second, on HiFi removing the sync term returns it to the baseline (0.4834→0.4735), so the sync term is not the HiFi culprit. Third, HiFi's softness comes from the long-horizon backbone itself importing DeepSpeak-specific temporal fingerprints that reverse on joint generators. In short, the component that transfers on DFE-2024 is the same one that fails on HiFi: a partial result, not a net win.

The bounded positive must be read as carefully as the negatives. **0.5796 on DFE-2024 is not a detection solution:** it sits below the 0.65 transfer gate and below any threshold at which an operator would deploy a screenshot detector with confidence; its value is comparative and attributive rather than operational. It is the strongest score any recipe we ran reaches on a large in-the-wild benchmark, it is produced on a single 24 GB GPU from a shallow long-horizon adapter on frozen features, and the gain is the one outcome that survives the sign-stability check that every other recipe fails, even though it does not cross the magnitude gate. That combination, gain-without-invariant, is why we label it a bounded positive rather than a fix: the frozen features still carry *some* sample-transferable signal worth exposing (long-horizon content regularity), while the cross-modal consistency axis itself does not transfer in this regime. Conflating the two (the content backbone, which helps, and the consistency prior, which does not) would misread the number as support for a hypothesis that remains falsified here.

E. Window-Alignment: A Falsified Data-Layer Hypothesis

Before blaming the model, we checked whether cross-domain caches were temporally misaligned. An audit found that the training-domain preprocessing clipped a centred 4 s window for both audio and video, whereas the cross-domain preprocessing downsampled full video but extracted only the *first* 4 s of audio, misaligning the axes for 74–99% of cross-domain clips. We fixed all cross-domain preprocessing to use the training-domain centred 4 s window, re-extracted, and re-evaluated the *same* LTCA checkpoint. HiFi moved by only +0.005 (0.4834→0.4887), both near random. Alignment was necessary for correctness but not the cause of the collapse: a concretely falsified prior, pinning the residual problem to the model layer.

F. End-to-End Alignment: A Seesaw, Not a Fix

The strongest remaining objection, echoing HAVIC/X-AVDT, is that consistency transfers only if features *co-evolve* with an alignment objective, and that the negative result is an artefact of freezing. We relaxed the freeze with a full fine-tune of the LCMA/LACA projections plus a per-frame alignment loss on authentic clips, $\lambda = 0.1$, 40 epochs (single-variable: same features, same Stage-1, same init), producing a textbook *seesaw*: HiFi is recovered (+0.026 →0.5097) only by forfeiting DFE-2024's hard-won gain (−0.045 →0.5351); FakeAV is unchanged (−0.001). Explicit alignment does not render consistency cross-generator invariant; it re-points the measure toward the training domain. This is the decisive negative signal: even co-evolution fails under single-source supervision. Figure 3 shows the seesaw directly: aligning recovers HiFi only by forfeiting DFE-2024, and the two moves are opposite in sign.

G. Reconstruction Consistency Prior: Reproduction Failure

One objection to the seesaw is that our alignment target was incomplete. We therefore rebuilt the fuller HAVIC-style prior: masked cross-modal reconstruction plus global semantic cross-reconstruction at Stage-1 on real data (60 epochs), then fine-tuned Stage-2 with the LTCA recipe. Logs confirm the terms entered backpropagation. The reproduction *fails*: DFE-2024 regresses by −0.050 (0.5585→0.5081) and HiFi moves only +0.005 (noise). A fully implemented prior still cannot produce a cross-generator invariant under the frozen, single-source regime: the fourth independent evidence line.

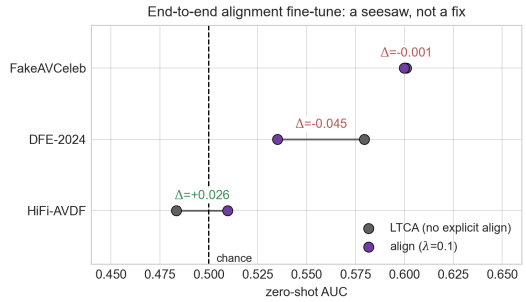


Fig. 3. End-to-end alignment fine-tune versus LTCA baseline: a seesaw. DFE-2024 drops ($\Delta = -0.045$) as HiFi recovers ($\Delta = +0.026$); FakeAV is unchanged.

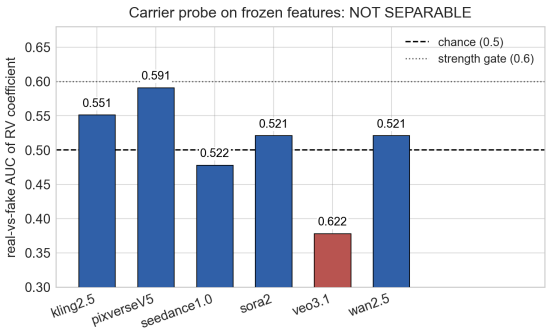


Fig. 4. Carrier probe on frozen features per HiFi-AVDF generator. Only veo3.1 exceeds the 0.6 strength gate, and it is sign-reversed; the verdict is NOT_SEPARABLE.

H. Multi-Source Prerequisite and Carrier Probe

a) Multi-source prerequisite fails.: Before committing GPU time to a multi-generator objective (a ROGA-style alignment), we tested its statistical precondition at zero training cost: whether the consistency signal is sign-stable across individual generators. Grouping HiFi's six commercial generators and measuring real-vs-fake per generator, `corr_pre` splits +3/−3 (for kling2.5 and wan2.5 real consistency is higher; for sora2/veo3.1/seedance1.0 it is lower), with strong separability in only one group, so even a multi-source objective would have no consistent target to push toward.

b) Carrier probe: NOT_SEPARABLE.: To make the attribution airtight, and to guide the remaining high-cost fork (a joint backbone such as AV-HuBERT), we ran a linear carrier probe built on centred kernel alignment (CKA) and the RV coefficient, the latter measuring the congruence of two covariance matrices, directly on the cached features (per generator as its own balanced task, 226–394 clips) to test whether the offline feature set contains any audio-visual covariance that is sign-stable and strong across generators. Only **1/6** generators attains strength ≥ 0.6 (veo3.1, and it is *sign-reversed*, Table III, Figure 4): the verdict is *NOT_SEPARABLE*. The frozen separate backbones expose no cross-generator consistency gradient whatsoever; the absence is structural, because wav2vec2 is pre-trained on single-modality audio self-supervision and never encodes audio-visual alignment. This explains why swapping in further separate backbones would also fail: the missing information is cross-modal structure, not per-modality capacity.

1) Not simply a data or protocol artefact: These failures might reflect a weak configuration rather than the hypothesis. Three checks argue against that reading. First, the failures are *consistent* across five independent evidence lines that share only the frozen feature set: a data-fidelity problem would show up in-domain too, yet in-domain performance is strong (AUC 0.9873 video-only) and the separable audio signal is separately confirmed (0.827). Second, the collapse has a concrete, reproducible signature: the carrier probe shows the audio–visual covariance is sign-*unstable* across generators (one generator sign-reversed, one group sign-split), and alignment fine-tune lifts one cross-domain set while degrading another (a seesaw),

TABLE III
REPRESENTATION-CARRIER PROBE: AUDIO–VISUAL COVARIANCE (RV COEFFICIENT) REAL-VS-FAKE PER HIFI-AVDF GENERATOR (STRENGTH = max(AUC, 1 − AUC)).

Generator	auc	strength
kling2.5	0.551	0.551
pixverseV5	0.591	0.591
seedance1.0	0.478	0.522
sora2	0.521	0.521
veo3.1	0.378	**0.622**
wan2.5	0.521	0.521

which is a representational phenomenon, not a training bug. Third, the one component that does transfer (LTCA) operates on a different axis (long-horizon content), and its gain is directionally stable on the primary benchmark. No single confound explains why several structurally different recipes fail identically; the explanation lives in the representation.

2) Why the null is not an artefact of a small evaluation: The natural objection is that more data or more generators could resurrect the signal. On our evidence this is unlikely, for quantitative reasons. First, the carrier probe is a *feature-level* scan of every cached clip rather than a single trained model: it reads the offline tensors directly, so its NOT_SEPARABLE verdict reflects the representation itself, not the per-configuration sample size. Second, the discriminating quantity is the *sign instability*, not the mean AUC: across the six commercial generators the audio–visual covariance is sign-reversed or sign-split, and no realistic increase in clip count can fix a sign that flips between generators; more data would only shrink the confidence interval around the same split. Third, the in-domain signal is strong (AUC 0.9873) on the identical pipeline, so a data-fidelity or implementation fault would have surfaced in-domain, where it did not. What additional generators would add is therefore not confidence in the consistency hypothesis but a test of the adjacent claim that *co-trained* or *real-prior* backbones (HAVIC-style, ROGA-style) behave differently: precisely the work we delegate outside our frozen, single-source scope.

TABLE IV
GEND COMPONENT ABLATION: CROSS-DOMAIN AUC, 3-SEED MEAN±STD. Δ IS RELATIVE TO THE FULL GEND PRINCIPAL CONFIGURATION.

Variant	FakeAV	DFE-2024	HiFi-AVDF
Principal (full GenD)	0.594 ± 0.024	0.529 ± 0.029	0.488 ± 0.001
w/o Rec B (λ=0)	0.558 ± 0.036	0.508 ± 0.026	0.469 ± 0.024
w/o Rec C (i.i.d.)	0.593 ± 0.042	0.516 ± 0.045	0.481 ± 0.018
Δ w/o Rec B	−0.036	−0.021	−0.019
Δ w/o Rec C	−0.001	−0.013	−0.008

a) A reusable single-variable attribution pipeline.: The procedure that produced these numbers is itself a contribution, stated generically so that it transfers to other detectors built under the same budget. It has three stages, ordered so that the cheapest evidence is gathered first. *(i) Shortcut scan.* Run the cheapest modality-identity and frequency/physical probes on the cached features before any training: each costs a few minutes on one GPU and localises whether the signal lives in supervision or in the frozen representation, which decides where the next experiment should spend its budget. *(ii) Modality budget.* Zero-mask each stream to quantify how much of the in-domain decision each modality carries, and linearly probe the raw embeddings to check whether an unexploited signal exists but is not wired into the supervised head: an asymmetry we found (separable audio, 0.827, yet random in the fused model) that a training-only analysis would have missed. *(iii) Carrier probe with a sign gate.* Before committing GPU time to any expensive co-training or multi-source objective, verify at feature level that the target covariance is sign-stable and strong across generators; if no generator satisfies the strength gate, the expensive recipe has no consistent target and should not be run. The discipline that makes the pipeline sound is the sign gate itself: it flags a direction that flips between generators as non-transferable even when a mean AUC looks non-random, which is precisely the trap the seesaw and the +3/−3 prerequisite result exposed. The ordering (cheapest before costly, feature-level before trainable, sign before magnitude) lets a resource-constrained lab avoid spending a GPU budget on a recipe its own features cannot justify.

I. GenD Component Ablation: Trend-Level, Not Significant

To attribute each GenD component independently, we switch off one structural regulariser at a time, either the hyperspherical metric (Rec B, $\lambda_{\text{align}} = \lambda_{\text{uniform}} = 0$) or the same-source pairing (Rec C, falling back to i.i.d. sampling), while keeping the architecture and full recipe identical (LTCA, batch 512, LR 1×10^{-4}, 80-epoch early stopping). Each variant is trained with 3 seeds and measured as mean±std (Table IV, Figure 5).

a) Rec B carries the main (trend-level) contribution.: Removing it lowers all three sets ($\Delta \in [-0.036, -0.019]$), the largest on FakeAVCeleb (−0.036), consistent with the alignment and uniformity regulariser helping separate real/fake on the sphere. But its magnitude is below the per-domain seed deviation ($\sigma \approx 0.024$–0.036), so it is *not* statistically significant: trend-level evidence only. **Rec C contributes essentially nothing** (−0.001 on FakeAV, −0.008 to −0.013 elsewhere). All variants converge to the same 0.958–0.983 in-domain band, ruling out a convergence confound. This corroborates the mechanism at component granularity: even the cross-domain mechanism we favoured most yields no significant gain; w/o-Rec-B seed 0 reaches the *highest* in-domain accuracy (0.9829) yet the worst cross-domain result (FakeAV 0.5172): in-domain accuracy does not transfer.

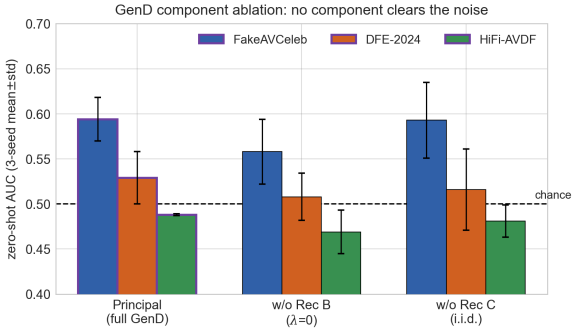


Fig. 5. GenD component ablation (3-seed mean ± std). No component moves any cross-domain set beyond its seed deviation; error bars overlap chance.

TABLE V
MODULE ABLATION: IN-DOMAIN BEST VALIDATION AUC AND CROSS-DOMAIN AUC, SINGLE SEED. THE V2 ROW IS THE PRINCIPAL CONFIGURATION.

Variant	In-domain	FakeAV	DFE-2024	HiFi-AVDF
v2 (full)	0.9758	0.6187	0.5350	0.4735
w/o LCMA	**0.7457**	**0.4909**	**0.4757**	**0.4570**
w/o DWF	0.9827	0.6287	0.5145	0.4777
LACA-corr	0.9839	0.6194	0.5171	0.4870
LACA-biomech	0.9845	0.6229	0.5207	0.4784

J. Module Ablation: LCMA Carries the Discriminative Load

To localise the collapse inside the adapter rather than in the training recipe, we ablate the three trainable modules one at a time on identical features, identical Stage-1 initialisation, batch size and schedule (80 epochs, early stopping). Every checkpoint is scored on the identical held-out caches (FakeAVCeleb 19,725 clips, Deepfake-Eval-2024 728, HiFi-AVDF 3,787). The three variants remove LCMA (`-disable_lcma`), replace the learnable DWF gate with fixed 0.5/0.5 fusion (`-disable_dwf`), and isolate each LACA branch (`laca_mode` ∈ {`corr`,`biomech`}). In-domain, only the first is costly: removing LCMA drops the best validation AUC from 0.9758 to **0.7457**, while the other three variants remain in the 0.98 band (Table V).

The cross-domain pattern is sharper than the in-domain one. Removing LCMA does not merely weaken the model: all three held-out sets fall *below* chance (0.4909/0.4757/0.4570), so the ranking *inverts* rather than attenuates. That is the signature of a generator-specific fingerprint—the decision learned on DeepSpeak reverses sign on every held-out generator family—which is the same phenomenon the sign gate of §V detects at feature level, now observed as an end-to-end behavioural

consequence. The other three variants sit within 0.021 of the v2 baseline, inside the between-seed standard deviation of the principal configuration (σ = 0.009–0.022 across the three sets), so we cannot resolve their individual contribution at this experimental resolution and therefore claim neither benefit nor harm for the DWF gate or for either LACA branch. One caveat is required: because $\mathcal{L}_{\text{shared}}$ is defined on the corr branch, isolating the biomech branch also removes that term from the objective, so the LACA-biomech row reflects the loss of both the branch and its alignment term rather than the branch alone. A second caveat bounds the interpretation: removing LCMA also removes the only cross-modal fusion path, so the inversion cannot be attributed to the consistency prior in isolation; what the result establishes is that the adapter, rather than the training recipe, carries the part of the decision that transfers at all.

K. Mechanism

Five independent lines, (1) the audio-supervision bottleneck with a separable-but-unused audio probe (0.827), (2) the seesaw alignment negative transfer, (3) the reconstruction-prior reproduction failure, (4) the multi-source prerequisite failure (+3/−3), and (5) the NOT_SEPARABLE carrier probe, converge on a causal chain from representation to outcome. CLIP is trained on image–text pairs and wav2vec2 on raw waveforms; neither objective ever paired lip motion with its acoustic consequence, so neither feature space contains a direction that encodes that coupling. Offline caching removes the one mechanism that could repair this: the encoders are never updated, and no downstream objective can write the missing cross-modal direction into them. A single training source compounds the effect, as any residual cross-modal signal is absorbed into generator-specific statistics rather than learned as a general cue. The carrier probe measures the consequence directly, while the pre-training objective explains the cause; the two agree.

> *Under the offline-frozen, single-source-supervised paradigm, audio-visual consistency never becomes a cross-generator invariant. Every convenient signature examined (landmark jitter, autocorrelation, coherence scalar, alignment target, reconstruction prior) is a training-domain (DeepSpeak) fingerprint: strong in-domain, sign-reversing or vanishing across generators. The model's in-domain 0.9758 is built from these generator-specific fingerprints superimposed; cross-domain they disappear, and the residual separable audio content signal (0.827) is never wired into the supervised decision.*

Two observations follow. First, this is *domain overfitting / shortcut learning*, not classical overfitting: training-domain augmentation cannot fix it. Second, the locus is the *feature layer*: changing supervision, architecture, alignment, or prior all fail because none can manufacture a cross-generator invariant that the frozen separate features do not contain.

L. What Is Falsified, and What Is Not

What is falsified is the conjunction

$$\begin{aligned}&(\text{frozen independent backbones}) \wedge (\text{single training source})\\ &\wedge (\text{supervised real/fake head}) \Rightarrow \text{consistency}\\ &\text{as a cross-generator invariant.}\end{aligned}$$

It does *not* falsify the consistency direction: HAVIC and X-AVDT transfer precisely because their representations co-evolve with an alignment objective from scale-appropriate pre-training and/or exploit white-box generator access, both outside our offline-frozen, black-box, single-GPU regime. The negative result concerns the *feature framework*; the field's strongest methods operate outside it. Where a lightweight, frozen, black-box detector can still help is bounded but real: the LTCA result (DFE-2024 0.5796) demonstrates a transferable gain from long-horizon content consistency, and the diagnostic methodology that localised it is reusable.

M. Limitations and Threats to Validity

We enumerate the principal threats to the validity of the falsification and how each is bounded.

a) Single training source.: Sync-X is trained on precisely one corpus (DeepSpeak v1.1), so the finding "consistency does not transfer" is established for a single-source, offline-frozen regime rather than for cross-modal consistency in general. This is deliberate: it isolates the supervision axis, and the same conclusion is independently reproduced under variant objectives (reconstruction prior, alignment fine-tune) rather than resting on one configuration. It does, however, mean we cannot extrapolate to multi-generator or co-trained settings: those are exactly the regimes where HAVIC and X-AVDT operate and where we make no claim.

b) Black-box, frozen design.: We never co-train the heavy encoders and never access generator internals. Consequently the carrier-probe verdict (NOT_SEPARABLE) applies to the *separate-frozen-feature* representation we evaluate, and says nothing about whether a co-trained joint backbone would expose usable covariance. We present the carrier probe as the mechanism *within* our regime, not as a property of all audio–visual representations.

c) No external-baseline reproduction.: We do not reproduce HAVIC or X-AVDT, and we make no claim to outperform them. Three conditions of those recipes are unavailable here rather than overlooked: they co-train the audio-visual backbones, whereas our pipeline trains on offline features whose encoders never re-enter the graph; they rely on scale-appropriate multi-source pre-training, whereas we hold a single supervised source of 10,550 clips; and their reported numbers use their own splits and preprocessing, so a cross-protocol comparison against our FakeAVCeleb/DFE-2024/HiFi-AVDF caches would be uninformative. The budget that a reproduction would consume is instead spent on the feature-level carrier probe, which decides whether a joint-backbone retrain is worth committing to at all. Accordingly we report no head-to-head comparison against their published state-of-the-art numbers:

under a different training protocol such a table would compare protocols rather than methods, and the claim advanced here is an attributed negative result, not a leaderboard position.

d) Evaluation scale and statistical conventions.: Each cross-domain set is evaluated at clip level on a limited number of clips per generator (226–394 per generator in the carrier probe), effect sizes are small and seed noise is non-negligible ($\sigma \approx 0.024$–0.045 for the GenD ablation; the principal configuration's three seeds give ±0.0222, ±0.0181 and ±0.0086 on FakeAVCeleb, DFE-2024 and HiFi-AVDF respectively). That carrier sample is bounded by what the dataset provides rather than by our subsampling: each HiFi-AVDF generator offers 226–394 usable clips, so the per-class cap of 600/400 was never active, and re-running the probe with it raised to 3000/3000 reproduces the per-generator statistics and the NOT_SEPARABLE verdict exactly. We therefore report multi-seed means only where a numerical delta is at stake and deliberately refrain from claiming statistical significance where the seed bounds overlap (as in the GenD and module ablations, where every non-LCMA delta falls within the seed deviation); in-domain-versus-cross-domain comparisons mix single-seed (mechanism) and three-seed (delta) measurements, and this asymmetry is disclosed explicitly rather than hidden. No confidence interval is fabricated; where a value is not available it is marked $--$.

Reporting the finding with its variance, gates, and excluded scopes is part of the methodological contribution: a reader can see where the claim holds and where it does not.

N. Implications for the Current Generator Landscape

The generators that erased the first-generation artefact cues (Seedance 2.0, SkyReels V4, the Sora 2 family) are the native-joint engines that tie speech to generated mouths in a single pass, and they are the most realistic threat to the audio–visual consistency assumption. A detector that freezes off-the-shelf unimodal backbones cannot outsource the defence to a consistency prior those backbones were never trained to represent; on our evidence the coupling has to be learned by a backbone that sees both modalities together and co-evolves on authentic data, the HAVIC-style direction we do not claim. Boundedly, the LTCA result shows that long-horizon content regularity is worth preserving even in a frozen design. Both implications follow from one claim: cross-modal consistency is real but is not a free, offline, cross-generator property.

These findings also bound what an evaluator should trust from a frozen-backbone detector. A high in-domain AUC on a single generator is not evidence of transferability: the same checkpoint that reports 0.9758 in-domain sits at 0.4735–0.535 across three held-out sets, and the sign of its most convenient cues reverses between generators. A deployment claim should therefore report cross-domain AUC with the generator names stated, a sign gate applied, and the overlap between training and evaluation populations made explicit, the reporting convention we adopt throughout.

VI. Conclusion

Under a resource-constrained, frozen-backbone, single-source design, the consistency-first hypothesis yields a negative result with one bounded positive. Cross-domain robustness does not survive the train/test shift: in-domain strength is visual (video-only 0.9873) while audio is near random (0.5197) despite separable features (0.827); three training recipes on identical features fail; alignment correction is inert; and the carrier probe is NOT_SEPARABLE. Consistency, as captured here, is a DeepSpeak-specific fingerprint rather than a cross-generator invariant, and the only transferable component (LTCA, 0.5796 on Deepfake-Eval-2024) is a long-horizon content backbone that still degrades HiFi-AVDF. The result reframes the success-framed literature: the gain of joint-backbone systems is carried by the representation, not by a consistency loss that can be added on top of frozen features.

These conclusions are bounded by one training source, a frozen design that never co-trains backbones, and clip-level metrics on three cross-domain sets.

a) A falsifiable next experiment.: Repeating the identical probe on a co-evolved joint backbone (AV-HuBERT-style, aligned pre-training) would revert the NOT_SEPARABLE verdict only if sign-stable covariance with strength ≥ 0.6 emerges on at least five of six HiFi-AVDF generators; because cropping, windowing, probes, gates and the head are reused verbatim, any change is attributable to the representation alone.

The experiments are reproducible, and the single-variable pipeline can be re-run in settings where the consistency hypothesis can succeed.

Statement of Contributions and Acknowledgements

In accordance with the assessment requirements, this section distinguishes the author's own work from that of others. The student personally designed the Sync-X framework and its single-variable attribution protocol, wrote all preprocessing, training, probe, and analysis code, ran every experiment and ablation reported, chose all configuration hyperparameters, and drafted the entire text of this research paper. The supervisor, Asad Malik, provided guidance on study design and comments on drafts but neither wrote the paper nor the experimental code.

The following external resources were used and are credited explicitly so that no work of others is overclaimed. (i) *Datasets* were collected and released by third parties (DeepSpeak v1.1 by Barrington et al., VoxCeleb2 by Chung et al., FakeAVCeleb by Khalid et al., Deepfake-Eval-2024 by Chandra et al., HiFi-AVDF by Peng et al., and FaceForensics++ by Rössler et al.), each under its stated licence, and none is claimed as collected for this project. (ii) *Frozen backbones* CLIP ViT-B/16 and wav2vec2-base-960h are publicly released models by Radford et al. and Baevski et al., loaded as drop-in PyTorch modules without modification. (iii) *MediaPipe FaceMesh* is Google's open-source library used as-is for landmark extraction. (iv) *The GenD-style objectives* (hyperspherical alignment/uniformity and same-source pairing) are drawn from the cited method of Yermakov et al. and were integrated into the Stage-2

loss and ablated by the student. Reproducibility details, code listings, and configuration accompany this thesis submission as supporting material.

This research paper is written afresh and does not duplicate the previously assessed literature review (Part 1 of the thesis); the overlap check between the two is limited to the shared reference lists, and no substantive prose is repeated.

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